

4 - SIEM Deployment for Threats Detection

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Course: Forensics

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Date of submission: 19.04.2026

Task 1 — Wazuh Deployment

1.1 Deployment Scenario

- Chosen deployment method:

In this lab I used All-in-One deployment (server-wazuh components are hosted on one VM), you can divide hosting all main components server, indexer, dashboard between different VMs, containers.

- Environment location (local / cloud):

- Local network via VMs

- Reason for choice:

- Safe, cheap, controllable, easy to deploy.

1.2 Lab Environment Nodes

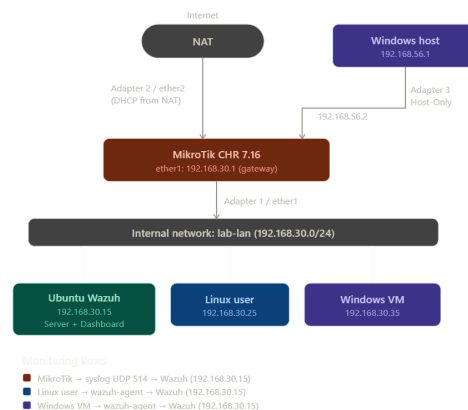


Figure — Infrastructure diagram

- Network topology used (NAT, HOST-ONLY, INTERNAL):

VM/Service	VirtualBox adapter(s)	IP	Info
MikroTik CHR 7.16	Adapter 1: Internal Network lab-lan , Adapter 2: NAT Adapter 3: Host-Only Adapter → VirtualBox Host-Only Ethernet Adapter #1 (192.168.56.x)	ether1: DHCP (10.0.2.15), ether2: 192.168.30.1 ether2: 192.168.56.2	admin:admin
Ubuntu Wazuh	Adapter 1: Internal Network lab-lan	192.168.30.15 (wazuh)	ubuntu:ubuntu
Linux user	Adapter 1: Internal Network lab-lan	192.168.30.25	ubuntu:ubuntu

VM/Service	VirtualBox adapter(s)	IP	Info
Windows	Adapter 1: Internal Network lab-lan	192.168.30.35	user:user and admin:admin
Wazuh Dashboard			admin:9zq2AK79*8vOZ6ID?? +CHVDgcNMS31Sp
Linux Honey Pot Machine	Yandex Cloud machine		Created for testing some separate wazuh features like log rotations

- **Verification steps taken to confirm isolation:**

```
[admin@MikroTik] > /ip address add address=192.168.30.1/24 interface=ether1
[admin@MikroTik] > /ip dhcp-client add interface=ether2 disabled=no
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether2 action
=masquerade
[admin@MikroTik] > /ip dns set allow-remote-requests=yes
[admin@MikroTik] > /ping 8.8.8.8

  SEQ HOST                                SIZE TTL TIME          STATUS
  --  --  --
0 8.8.8.8                                56 255 16ms201us
1 8.8.8.8                                56 255 1ms339us
2 8.8.8.8                                56 255 1ms58us
sent=3 received=3 packet-loss=0% min-rtt=1ms58us avg-rtt=6ms199us
max-rtt=16ms201us

[admin@MikroTik] > /ping google.com

  SEQ HOST                                SIZE TTL TIME          STATUS
  --  --  --
0 216.239.38.120                         56 255 1ms20us
1 216.239.38.120                         56 255 972us
2 216.239.38.120                         56 255 1ms195us
sent=3 received=3 packet-loss=0% min-rtt=972us avg-rtt=1ms62us
max-rtt=1ms195us
```

Figure — Mikrotik communication setup and test via **ping**

```
Linux-machine [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@linux-machine:~# cat /etc/netplan/01-static.yaml
network:
  version: 2
  ethernet:
    enp0s3:
      addresses: [192.168.30.25/24]
      routes:
        - to: default
          via: 192.168.30.1
      nameservers:
        addresses: [192.168.30.1, 8.8.8.8]
root@linux-machine:~# ping 192.168.30.1
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data.
64 bytes from 192.168.30.1: icmp_seq=1 ttl=64 time=18.5 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=64 time=6.88 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=64 time=13.3 ms
^C
--- 192.168.30.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 6.878/12.293/18.529/4.764 ms
root@linux-machine:~# ping 192.168.30.15
PING 192.168.30.15 (192.168.30.15) 56(84) bytes of data.
64 bytes from 192.168.30.15: icmp_seq=1 ttl=64 time=3.05 ms
64 bytes from 192.168.30.15: icmp_seq=2 ttl=64 time=1.18 ms
64 bytes from 192.168.30.15: icmp_seq=3 ttl=64 time=1.17 ms
^C
--- 192.168.30.15 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.172/1.799/3.046/0.881 ms
root@linux-machine:~# ping google.com
PING google.com (74.125.205.138) 56(84) bytes of data.
64 bytes from 1e-in-f138.1e100.net (74.125.205.138): icmp_seq=1 ttl=254 time=20.0 ms
64 bytes from 1e-in-f138.1e100.net (74.125.205.138): icmp_seq=2 ttl=254 time=62.2 ms
64 bytes from 1e-in-f138.1e100.net (74.125.205.138): icmp_seq=3 ttl=254 time=42.6 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 20.042/41.608/62.184/17.218 ms
root@linux-machine:~#
```

Figure — Linux machine **ping** test (windows, wazuh, router, nat)

```
Ubuntu Controller (Clean Ubuntu Desktop) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 22 15:07

root@UbuntuDesktop: ~
root@UbuntuDesktop:~# cat /etc/netplan/01-static.yaml
network:
  version: 2
  ethernets:
    enp0s3:
      addresses: [192.168.30.15/24]
      routes:
        - to: default
          via: 192.168.30.1/24
      nameservers:
        addresses: [192.168.30.1, 8.8.8.8]
root@UbuntuDesktop:~# ping 192.168.30.1
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data:
64 bytes from 192.168.30.1: icmp_seq=1 ttl=64 time=18.2 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=64 time=7.24 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=64 time=3.89 ms
^C
--- 192.168.30.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 3.887/9.771/18.190/6.108 ms
root@UbuntuDesktop:~# ping google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data:
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=
8 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=
3 ms
^C
--- forcesafesearch.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 2.181/14.219/26.257/12.038 ms
root@UbuntuDesktop:~#
```

Figure — Wazuh machine `ping` test (windows,linux machine, router, nat)

```
Windows10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Select Administrator: Windows PowerShell

PS C:\Windows\system32> New-NetIPAddress -InterfaceIndex 4 -IPAddress 192.168.30.35 -PrefixLength 24 -DefaultGateway 192.168.30.1

IPAddress : 192.168.30.35
InterfaceIndex : 4
InterfaceAlias : Ethernet
AddressFamily : IPv4
Type : Unicast
PrefixLength : 24
PrefixOrigin : Manual
SuffixOrigin : Manual
AddressState : Tentative
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipSource : False
PolicyStore : ActiveStore

IPAddress : 192.168.30.35
InterfaceIndex : 4
InterfaceAlias : Ethernet
AddressFamily : IPv4
Type : Unicast
PrefixLength : 24
PrefixOrigin : Manual
SuffixOrigin : Manual
AddressState : Invalid
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipSource : False
PolicyStore : PersistentStore
```

Figure — Windows machine setting up gateway


```

# ether2 = NAT adapter (internet via VirtualBox NAT)
/ip dhcp-client add interface=ether2 disabled=no

# Enable NAT masquerade so LAN hosts reach internet
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade

# Allow DNS relay for LAN clients
/ip dns set allow-remote-requests=yes

# Verify internet works
/ping 8.8.8.8 count=3

# This puts Mikrotik on the same subnet as your Windows Host-Only adapter
/ip address add address=192.168.56.2/24 interface=ether3

# allow forwarding between ether3(With Host) and ether1 (Lan)
/ip firewall filter add chain=forward in-interface=ether3 out-interface=ether1 action=accept
/ip firewall filter add chain=forward in-interface=ether1 out-interface=ether3 action=accept

# On host machine Windows:
route add 192.168.30.0 mask 255.255.255.0 192.168.56.2 metric 5

# Ubuntu Wazuh server

# Find your interface name
ip link show

# Set static IP via netplan
sudo tee /etc/netplan/01-static.yaml << 'EOF'
network:
  version: 2
  ethernets:
    enp0s3:
      addresses: [192.168.30.15/24]
      routes:
        - to: default
          via: 192.168.30.1
      nameservers:
        addresses: [192.168.30.1, 8.8.8.8]
EOF

sudo netplan apply

# Verify
ping -c 3 192.168.30.1    # gateway
ping -c 3 8.8.8.8        # internet
ping -c 3 google.com     # DNS

# Linux user machine

```

```

# Same approach, different IP
sudo tee /etc/netplan/01-static.yaml << 'EOF'
network:
  version: 2
  ethernets:
    enp0s3:
      addresses: [192.168.30.25/24]
      routes:
        - to: default
          via: 192.168.30.1
      nameservers:
        addresses: [192.168.30.1, 8.8.8.8]
EOF

sudo netplan apply
ping -c 3 192.168.30.1
ping -c 3 192.168.30.15

# Windows VM

# Find your adapter index
Get-NetAdapter

# Found that it is 4

# Set static IP
New-NetIPAddress -InterfaceIndex 4 -IPAddress 192.168.30.35 -PrefixLength 24 -DefaultGateway 192.168.30.1
Set-DnsClientServerAddress -InterfaceIndex 4 -ServerAddresses 192.168.30.1,8.8.8.8

# Verify
ping 192.168.30.1
ping 192.168.30.15
ping 8.8.8.8
```


Replace `InterfaceIndex 6` with the actual index from `Get-NetAdapter`.

Step 6 – Verify full connectivity from every machine

Every VM should be able to reach every other VM and the internet:


```

ping 192.168.30.1      # MikroTik
ping 192.168.30.15    # Wazuh
ping 192.168.30.25    # Linux user
ping 192.168.30.35    # Windows
ping 8.8.8.8          # Internet

```


```

- Setting up basic Wazuh infrastructure

```

Install Wazuh agent on Linux user

#Install the following packages if missing:
apt-get install gnupg apt-transport-https

#Install the GPG key:
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gn
upg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg

#Add the repository:
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stab
le main" | tee -a /etc/apt/sources.list.d/wazuh.list

#Update the package information:
apt-get update

Select your package manager and run the command below. Replace the WAZUH_MANAGER value wit
h your Wazuh manager IP address or hostname:
WAZUH_MANAGER="192.168.30.15" apt-get install wazuh-agent

Enable and start the Wazuh agent service.
systemctl daemon-reload
systemctl enable wazuh-agent
systemctl start wazuh-agent

Install Wazuh agent on Windows

Download the Windows installer to start the installation process.
https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.4-1.msi

run installation (Powershell)
.\wazuh-agent-4.14.4-1.msi /q WAZUH_MANAGER="192.168.30.15"

Start (Powershell)
Start-Service wazuhsvc

Files are stored C:\Program Files (x86)\ossec-agent

Send all logs to Wazuh via syslog (UDP 514)
/system logging action set remote address=192.168.30.15 remote-port=514

/system logging add topics=info action=remote
/system logging add topics=error action=remote
/system logging add topics=warning action=remote
/system logging add topics=critical action=remote
/system logging add topics=firewall action=remote

```

## Task 2 — Configure the SIEM

## Configuring: MikroTik to send syslog to Wazuh Server

- Paste the following code snippet at the bottom of the `/etc/rsyslog.conf` file on the Ubuntu endpoint. This code enables the UDP port **514** to listen for syslogs and adds a location to store the security events. Replaced `<YOUR_MIKROTIK_IP_ADDRESS>` with my IP address of the MikroTik device.

```
root@Linux-Wazuh:/media# tail -n 7 /etc/rsyslog.conf
$IncludeConfig /etc/rsyslog.d/*.conf

$ModLoad imudp
$UDPServerRun 514

if $fromhost-ip startsWith '192.168.30.1' then /var/log/mikrotik.log
~
root@Linux-Wazuh:/media#
```

```
root@Linux-Wazuh:/media# touch /var/log/mikrotik.log
root@Linux-Wazuh:/media# ls -l /var/log/mikrotik.log
-rw-rw-r-- 1 root root 0 Oct 10 10:10 /var/log/mikrotik.log
root@Linux-Wazuh:/media# systemctl restart rsyslog
root@Linux-Wazuh:/media# systemctl status rsyslog
```

Figure — `/etc/rsyslog.conf` file with new configurations and log file is created

- Create the `mikrotik.log` file in the `/var/log` directory to store the syslog events.
- Change the file ownership for the `/var/log/mikrotik.log` file to syslog and group to `adm`.
- Restart the rsyslog utility for the changes to take effect.

```
root@Linux-Wazuh:~# touch /var/log/mikrotik.log
root@Linux-Wazuh:~# chown syslog:adm /var/log/mikrotik.log
root@Linux-Wazuh:~# systemctl restart rsyslog
```

- Install WinBox on my host machine and connect to edit Mikrotik configurations.
  - SW already found to which endpoint to connect

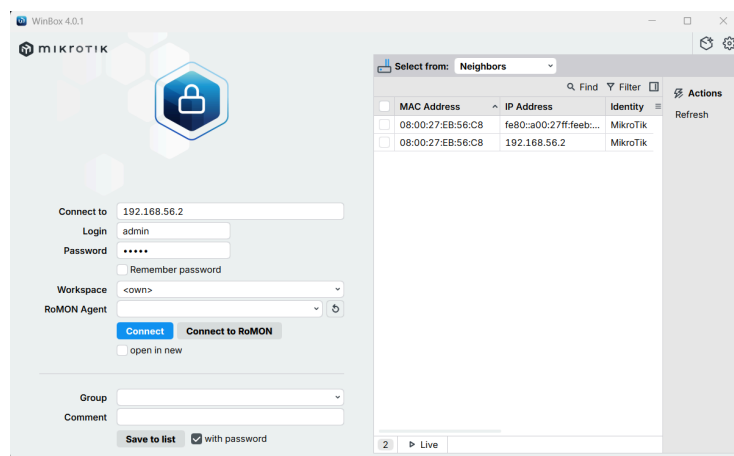


Figure — WinBox logging in to mikrotik router

- Open Logging tab

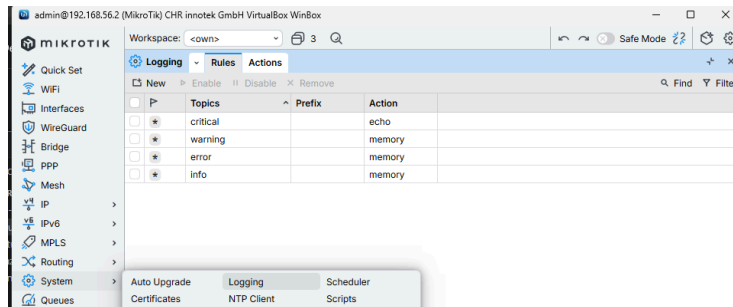


Figure — Inside mikrotik configuration for allowing forwarding to remote node

- Click on the **System > Logging > Rules** tab and create new or modify the existing rules where the topic's **error**, **info**, and **warning** are set to **remote**.

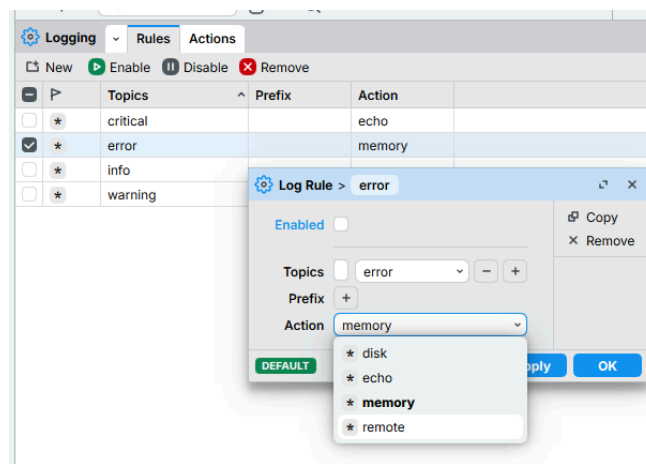


Figure — Changing rules to remote

| Logging                             |          |        |        |
|-------------------------------------|----------|--------|--------|
| Rules                               |          |        |        |
| Actions                             |          |        |        |
| New Enable Disable Remove           |          |        |        |
|                                     | Topics   | Prefix | Action |
| <input type="checkbox"/>            | critical |        | echo   |
| <input checked="" type="checkbox"/> | error    |        | remote |
| <input type="checkbox"/>            | info     |        | remote |
| <input type="checkbox"/>            | warning  |        | remote |

Figure — All rules are changed to remote

- Click the **System > Logging > Actions** tab to edit the **remote** action. Set the **Remote address** field to the IP address of the Wazuh manager and the **Remote port** field to the Rsyslog listening port.

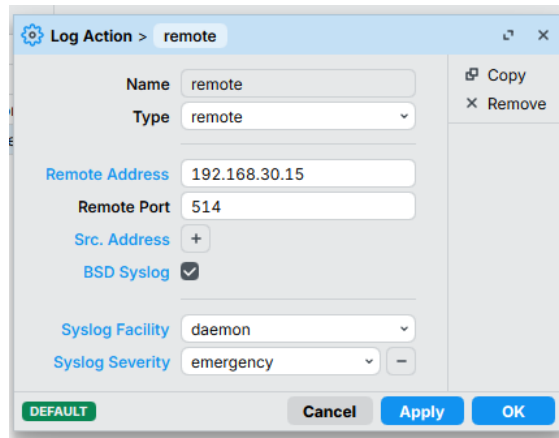


Figure — configuring remote action to actually point to my Wazuh Server

## Configuring: ossec

- On the Wazuh server, edit `/var/ossec/etc/ossec.conf` and add inside the `<ossec_config>` block:
  - `allowed-ips` specifies which endpoints are allowed to send logs to this server. Replace `<CIDR_NOTATION>` with the CIDR address of the network on which your endpoints are. If you monitor just one endpoint, you can put its IP address.
  - `local-ip` specifies the local IP address of the Wazuh server to be used for this connection. Replace `<WAZUH_MANAGER_IP>` with the IP address of the Wazuh server.

```
<remote>
 <connection>syslog</connection>
 <port>514</port>
 <protocol>udp</protocol>
 <allowed-ips>192.168.30.1</allowed-ips>
 <local_ip>192.168.30.15</local_ip>
</remote>
```

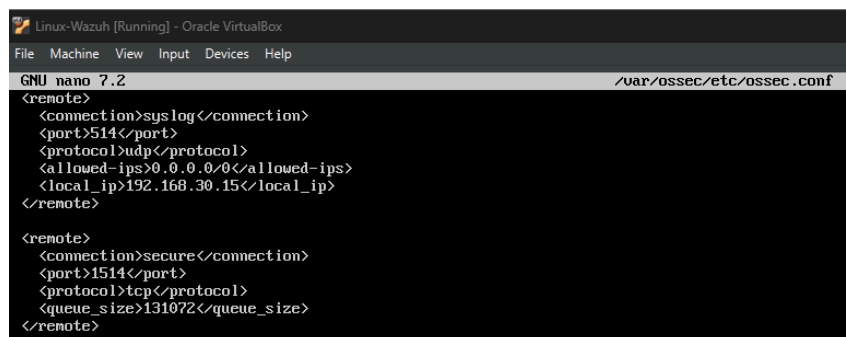


Figure — `/var/ossec/etc/ossec.conf` new block is added

- Setup mikrotik to allow WinBox UI

```
[admin@MikroTik] > /ip service enable winbox
[admin@MikroTik] > /ip firewall filter add chain=input protocol=tcp dst-port=829
1 action=accept place-before=0
```

## Configuring: decoders and rules

- Setting up Wazuh Decoders from [documentation](#).

- Wrote 2 files, send them to wazuh server
- Restart `sudo systemctl restart wazuh-manager`
- Content of the files:

`/var/ossec/etc/decoders/mikrotik_decoders.xml`

```
<decoder name="mikrotik">
 <prematch>^Router057.1-logs: </prematch>
</decoder>

<decoder name="mikrotik1">
 <parent>mikrotik</parent>
 <regex type="pcre2">\S+ (\w+ \d+ \d+:\d+:\d+) MikroTik user (\S+) (.*) from (\d+.\d+.\d+.\d+) via (\w+)</regex>
 <order>logtimestamp, logged_user, action, ip_address, protocol</order>
</decoder>

<decoder name="mikrotik1">
 <parent>mikrotik</parent>
 <regex type="pcre2">\S+ (\w+ \d+ \d+:\d+:\d+) MikroTik dhcp-client on (\S+) (.*) address (\d+.\d+.\d+.\d+)</regex>
 <order>logtimestamp, interface, action, ip_address</order>
</decoder>

<decoder name="mikrotik1">
 <parent>mikrotik</parent>
 <regex type="pcre2">\S+ (\w+ \d+ \d+:\d+:\d+) MikroTik router (\S+)</regex>
 <order>logtimestamp, action</order>
</decoder>
```

`var/ossec/etc/rules/mikrotik_rules.xml`

```
<group name="Mikrotik,">

 <rule id="110000" level="0">
 <decoded_as>mikrotik</decoded_as>
 <description>Mikrotik-Event</description>
 </rule>

 <rule id="110001" level="5">
 <if_sid>110000</if_sid>
 <match>dhcp-client on ether</match>
 <description>MikroTik dhcp-client received an IP address $(ip_address)</description>
 </rule>

 <rule id="110002" level="5">
 <if_sid>110000</if_sid>
 <match>rebooted</match>
 <description>MikroTik router rebooted</description>
 </rule>

 <rule id="110003" level="5">
 <if_sid>110000</if_sid>
 <match>logged out from</match>
 <description>MikroTik user logged out via $(protocol)</description>
 </rule>

 <rule id="110004" level="5">
 <if_sid>110000</if_sid>
 <match>logged in from</match>
 <description>MikroTik user logged in from $(ip_address) via $(protocol)</description>
 </rule>

</group>
```

```
root@Linux-Wazuh:/media# cp mikrotik_decoders.xml /var/ossec/etc/decoders/
root@Linux-Wazuh:/media# cp mikrotik_rules.xml /var/ossec/etc/r
rootcheck/ rules/
root@Linux-Wazuh:/media# cp mikrotik_rules.xml /var/ossec/etc/rules/
```

Figure — Copying created rules and decoders files to wazuh server

- Custom Rules and Decoders are seen in the dashboard, after the reboot of wazuh-manager.

Rules (6)

From here you can manage your rules.

relative\_dirname=etc/rules

WQLCustom rules

ID ↑	Description	Groups	Regulatory compliance	Level	File	Path
100001	ssh: authentication failed from IP 1.1.1.1.	authentication_fail etc, local, syslog, sshd	PCI_DSS	5	local_rules.xml	etc/rules
110000	Mikrotik-Event	Mikrotik		0	mikrotik_rules.xml	etc/rules
110001	MikroTik dhcp-client received an IP address ip_address	Mikrotik		5	mikrotik_rules.xml	etc/rules
110002	MikroTik router rebooted	Mikrotik		5	mikrotik_rules.xml	etc/rules
110003	MikroTik user logged out via protocol	Mikrotik		5	mikrotik_rules.xml	etc/rules
110004	MikroTik user logged in from ip_address via protocol	Mikrotik		5	mikrotik_rules.xml	etc/rules

Rows per page: 10

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Figure — Verifying that custom rules are seen and applied in the SIEM architecture

Decoders (5)

From here you can manage your decoders.

relative\_dirname=etc/decoders

WQLCustom decoders

Name	Program name	Order	File ↑	Path
local_decoder_example	{ "pattern": "local_decoder_example" }		local_decoder.xml	etc/decoders
mikrotik			mikrotik_decoders.xml	etc/decoders
mikrotik1		logtimestamp, logged_user, action, ip_address, protocol	mikrotik_decoders.xml	etc/decoders
mikrotik1		logtimestamp, interface, action, ip_address	mikrotik_decoders.xml	etc/decoders
mikrotik1		logtimestamp, action	mikrotik_decoders.xml	etc/decoders

Rows per page: 10

<1>

Figure — Verifying that custom decoders are seen and applied in the SIEM architecture

Configuring: verifying

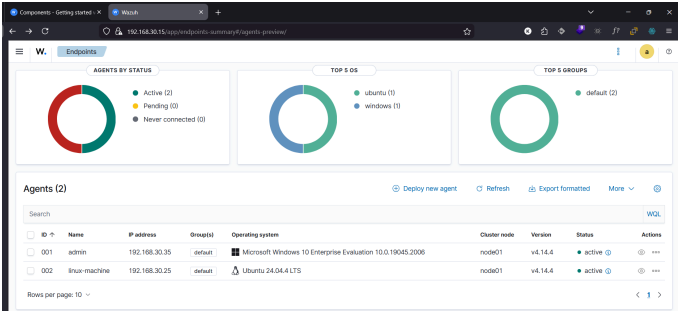


Figure — SIEM dashboard showing all connected agents/devices

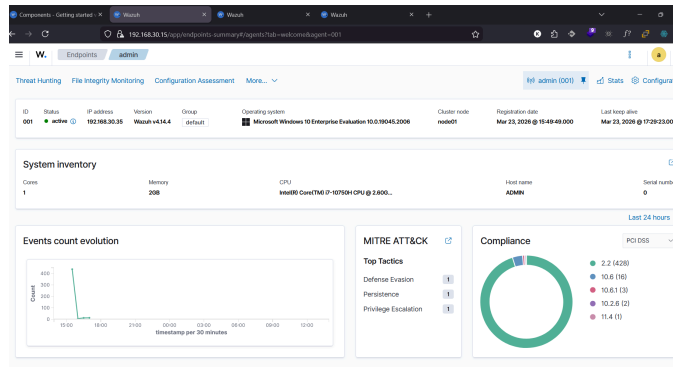


Figure — Log data visible from Device 2 (e.g. Windows)

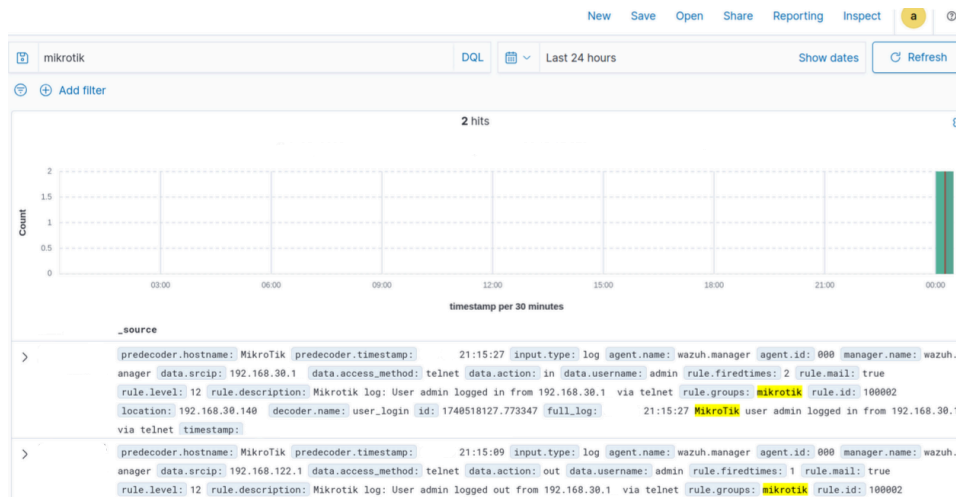


Figure — Log data visible from Device 3 (e.g. Network device)

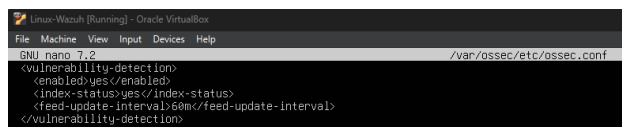
## 2.2 Features Enabled for Testing

- **Feature 1 (CVE detection and remediation):**
  - **ossec** configurations for **Vulnerability detection, time scanning**
  - Filebeat configuration
  - CVE vulnerable package
- **Feature 2 (Malware detection by SIEM):**
  - YARA integration: YARA detection rules and decoders, profiles (default and community sets), YARA Active Response scan (sh script)
  - **ossec** configuration connecting YARA to wazuh
  - Malware sample
- **Feature 3 (Brute-force attack simulation with attacker user blocking):**
  - SIEM rule triggering for brute force (Wazuh rule 5551, 5712, etc.)
  - **hydra** on attacker

## Task 3 — SIEM in Action

## 3.1 Feature 1 — CVE detection and remediation

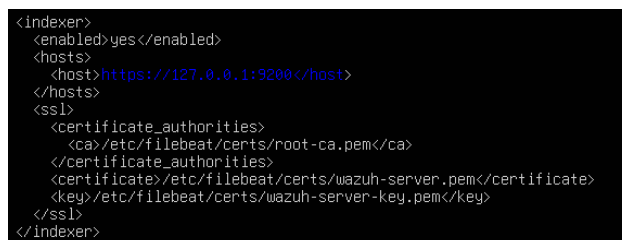
- Performed the following steps on the Wazuh server to confirm that the Wazuh Vulnerability Detection module is enabled and properly configured.
  - Check Vulnerability detection is enabled in Wazuh configuration `/var/ossec/etc/ossec.conf`



```
Linux-Wazuh [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 7.2 /var/ossec/etc/ossec.conf
<vulnerability-detection>
<enabled>yes</enabled>
<index-status>yes</index-status>
<feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>
```

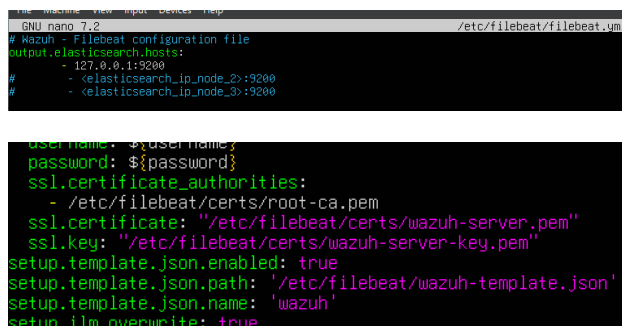
Figure — `/var/ossec/etc/ossec.conf`

- Check for host ip address and certificates match the ones inside `filebeat.yml`



```
<indexer>
<enabled>yes</enabled>
<hosts>
<host>https://127.0.0.1:9200</host>
</hosts>
<ssl>
<certificate_authorities>
<ca>/etc/filebeat/certs/root-ca.pem</ca>
</certificate_authorities>
<certificate>/etc/filebeat/certs/wazuh-server.pem</certificate>
<key>/etc/filebeat/certs/wazuh-server-key.pem</key>
</ssl>
</indexer>
```

Figure — `/var/ossec/etc/ossec.conf`

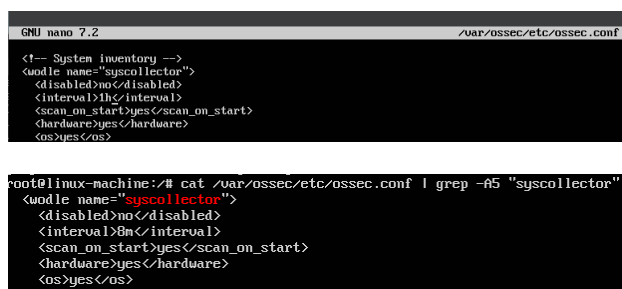


```
Linux-Wazuh [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 7.2 /etc/filebeat/filebeat.yml
Wazuh - Filebeat configuration file
output.elasticsearch:
 hosts:
 - 127.0.0.1:9200
 # - <elasticsearch_ip_node_2>:9200
 # - <elasticsearch_ip_node_3>:9200

user.name: ${user.name}
password: ${password}
ssl.certificate_authorities:
 - /etc/filebeat/certs/root-ca.pem
ssl.certificate: "/etc/filebeat/certs/wazuh-server.pem"
ssl.key: "/etc/filebeat/certs/wazuh-server-key.pem"
setup.template.json.enabled: true
setup.template.json.path: "/etc/filebeat/wazuh-template.json"
setup.template.json.name: "wazuh"
setup.ilm.overwrite: true
```

Figure — `/etc/filebeat/filebeat.yml`

- reduced time for scanning for vulnerabilities to 7 minutes in the `/var/ossec/etc/ossec.conf` file on the Wazuh agent



```
GNU nano 7.2 /var/ossec/etc/ossec.conf
<!-- System inventory -->
<module name="syscollector">
<disabled>no</disabled>
<interval>1h</interval>
<scan_on_start>yes</scan_on_start>
<hardware>yes</hardware>
<os>yes</os>

root@linux-machine:~# cat /var/ossec/etc/ossec.conf | grep -A5 "syscollector"
<module name="syscollector">
<disabled>no</disabled>
<interval>8m</interval>
<scan_on_start>yes</scan_on_start>
<hardware>yes</hardware>
<os>yes</os>
```

Figures — changing interval on Wazuh Agent configuration

- Go to <https://192.168.30.15/app/vulnerability-detection> and open tab Inventory too look at Vulnerable detections

- filtered them on my agent name
- Found 3 critical CVEs with `linux-image-6.8.0-106-generic`

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
linux-machine	linux-image-6.8.0-106-generic	6.8.0-106.106	IEEE P802.11-REVme D1.1 through D7.2 allows FragAttacks...	Critical	CVE-2025-27558
linux-machine	linux-image-6.8.0-106-generic	6.8.0-106.106	CWE-502 Deserialization of Untrusted Data at threeventmesh...	Critical	CVE-2024-56180
linux-machine	linux-image-6.8.0-106-generic	6.8.0-106.106	A flaw in netfilter could allow a network-connected attacker L...	Critical	CVE-2021-3773

Figures — 3 critical CVEs

- Removing this package

```
root@linux-machine:~# sudo apt remove linux-image-6.8.0-106-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
amd64-microcode bpfcc-tools bpftool fontconfig-config fonts-dejavu-core
libclang-cpp18 libclang1-18 libcrypt-dev libdbus-glib-1-2 libde265-0 lib
libheif-plugin-libde265 libheif1 libimobiledevice6 libjbig0 libjpeg-tur
libusbmuxd6 libwebp7 libxpm4 linux-headers-generic linux-libc-dev mamp
upower usbmuxd
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
linux-generic linux-image-6.8.0-106-generic linux-image-generic
0 upgraded, 0 newly installed, 3 to remove and 0 not upgraded.
After this operation, 15.1 MB disk space will be freed.
Do you want to continue? [Y/n] Y
```

Figures — removing critical package

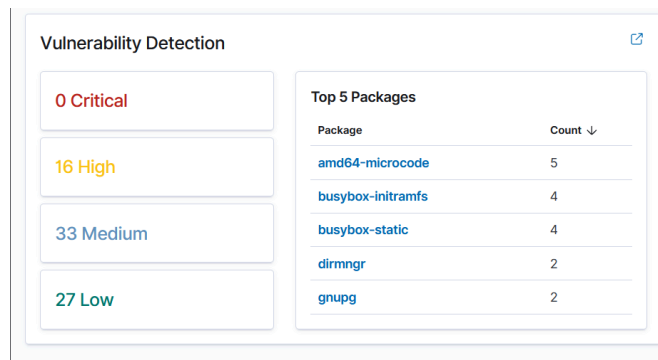


Figure — Remediation applied on Device 3

W. Vulnerability De... linux-machine

Dashboard Inventory Events

Search

wazuh.cluster.name: Linux-Wazuh agent.id: 002 Evaluated Under evaluation vulnerability.severity: Critical

No results match your search criteria

Figure — Updated scan showing vulnerability resolved on all devices

- Then to properly show how vulnerability would like when it is installed on running node, I decided to install this one package and check.

```
root@linux-machine:~# sudo apt install linux-image-6.8.0-106-generic
```

Figure — Installing vulnerable package (image)

- Now inside tab **Events** I can observe such vulnerable packet, that actualy causes a lot of CVEs of different severity.

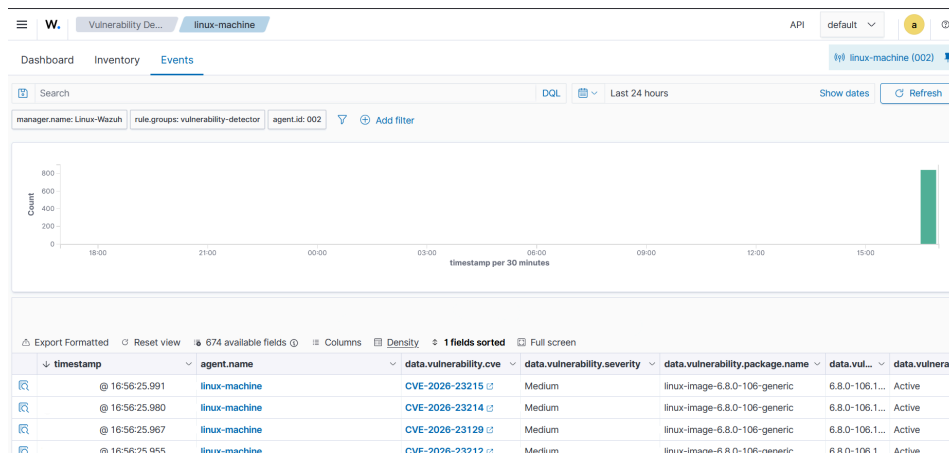


Figure — Events tab with targeted package

- Then I again uninstall it and was waiting to agent run again.

```
Removing linux-image-6.8.0-106-generic

You are running a kernel (version 6.8.0-106-generic) and attempting to remove the same vers
This can make the system unbootable as it will remove /boot/vmlinuz-6.8.0-106-generic and
/lib/modules/6.8.0-106-generic. This can only be fixed with a copy of the kernel image and
It is highly recommended to abort the kernel removal unless you are prepared to fix the sys
Abort kernel removal?

<Yes>

W: Removing the running kernel
W: Last kernel image has been removed, so removing the default symlinks
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Deleting /boot/initrd.img-6.8.0-106-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file '/etc/default/grub'
Generating grub configuration file ...
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
root@linux-machine:~#
```

Figure — vulnerable image is uninstalled

- Event now is Resolved:

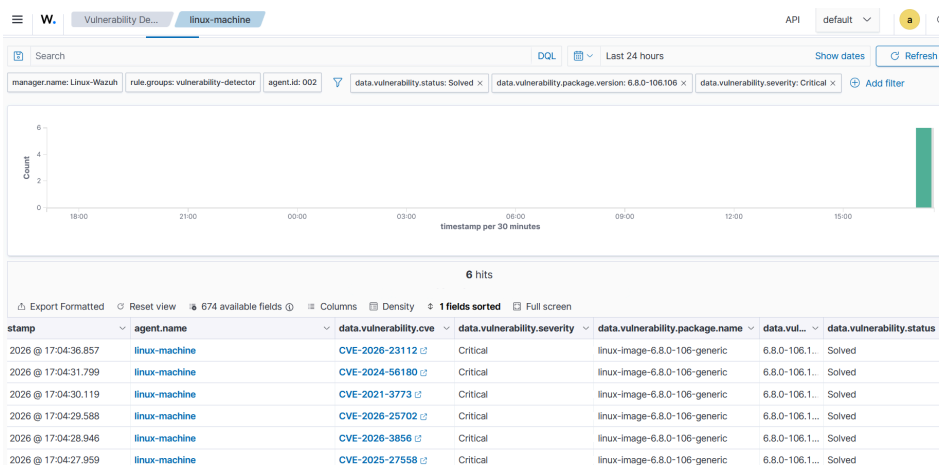


Figure — resolved status on all events with this package

## 3.2 Feature 2 — Malware detection by SIEM (WhisperGate Detection with YARA)

### Configuring linux-machine with agent

- Install on linux-machine YARA

```
sudo apt update
sudo apt install -y make gcc autoconf libtool libssl-dev pkg-config
sudo curl -LO https://github.com/VirusTotal/yara/archive/v4.5.5.tar.gz
sudo tar -xvzf v4.5.5.tar.gz -C /usr/local/bin/ && rm -f v4.5.5.tar.gz
cd /usr/local/bin/yara-4.5.5/
sudo ./bootstrap.sh && sudo ./configure && sudo make && sudo make install && sudo make check
```

- Test that YARA is running

```
make[2]: Entering directory '/usr/local/bin/yara-4.5.5'
make[3]: Entering directory '/usr/local/bin/yara-4.5.5'
PASS: test-arena
PASS: test-alignment
PASS: test-atoms
PASS: test-api
PASS: test-rules
PASS: test-pe
PASS: test-elf
PASS: test-version
PASS: test-bitmask
PASS: test-math
PASS: test-stack
PASS: test-re-split
PASS: test-async
PASS: test-string
PASS: test-exception
PASS: test-dotnet
=====
Testsuite summary for yara 4.5.5
=====
TOTAL: 16
PASS: 16
SKIP: 0
XFAIL: 0
FAIL: 0
XPASS: 0
ERROR: 0
=====
make[3]: Leaving directory '/usr/local/bin/yara-4.5.5'
make[2]: Leaving directory '/usr/local/bin/yara-4.5.5'
make[1]: Leaving directory '/usr/local/bin/yara-4.5.5'
root@linux-machine:~# yara
yara: wrong number of arguments
Usage: yara [OPTION]... [NAMESPACE:IRULES_FILE... FILE | DIR | PID]

Try `--help` for more options
```

- [illegible]

- ```
rule Whispergate_Stage_1 {
  meta:
    description = "Detects first stage payload from WhisperGate"
    author = "mmuir@cadosecurity.com"
    date = "2022-01-17"
    license = "Apache License 2.0"
    hash = "a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92"
  strings:
    $a = { 31 41 56 4E 4D 36 38 67 6A 36 50 47 50 46 63 4A 75 66 74 4B 41 54 61 34 57 4C 6E 7A 67 38 66 70 66 76 }
    $b = { 38 42 45 44 43 34 31 31 30 31 32 41 33 33 42 41 33 34 46 34 39 31 33 30 44 30 46 31 38 36 39 39 33 43 36 41 33 32 44
41 44 38 39 37 36 46 36 41 35 44 38 32 43 31 45 44 32 33 30 35 34 43 30 35 37 45 43 45 44 35 34 39 36 46 36 35 }
    $c = { 24 31 30 6B 20 76 69 61 20 62 69 74 63 6F 69 6E 20 77 61 6C 6C 65 74 }
    $d = { 74 6F 78 20 49 44 }
  condition:
    uint16(0) == 0x5A4D and all of them
}

rule Whispergate_Stage_2 {
  meta:
    description = "Detects second stage payload from WhisperGate"
    author = "mmuir@cadosecurity.com"
    date = "2022-01-17"
    license = "Apache License 2.0"
    hash = "dcbbae5alc61dbbbb7cd6dc5dd1eb1169f5329958d38b58c3fd9384081c9b78"
  strings:
    $a = { 6D 5F 49 6E 74 65 72 63 65 70 74 6F 72 }
    $b = { 6D 5F 62 31 36 65 37 33 65 30 64 61 61 63 34 62 34 33 62 36 35 36 36 39 30 31 62 35 34 32 34 63 35 33 }
    $c = { 6D 5F 34 33 37 37 33 32 63 65 65 35 66 35 34 64 37 64 38 34 61 64 64 37 62 64 33 30 39 37 64 33 63 61 }
    $d = { 6D 5F 30 64 62 39 37 30 38 63 66 36 34 39 34 30 38 32 39 66 39 61 66 38 37 65 64 65 65 64 66 36 30 65 }
    $e = { 6D 5F 65 31 34 33 33 31 36 38 32 30 62 31 34 64 30 33 38 38 61 37 32 37 34 34 33 38 65 63 30 37 38 64 }
    $f = { 6D 5F 66 33 31 30 39 30 63 37 31 35 64 65 34 62 30 62 61 62 64 33 31 61 36 33 34 31 31 30 34 36 63 38 }
    $g = { 6D 5F 36 31 31 64 31 61 62 63 33 32 66 63 34 66 64 38 61 33 34 65 30 34 34 66 39 37 33 34 34 31 64 61 }
    $h = { 6D 5F 37 37 34 62 39 32 31 30 64 39 38 31 34 32 65 62 62 34 31 33 35 35 39 64 61 61 65 35 61 34 34 }
  condition:
    uint16(0) == 0x5A4D and all of them
}
```

- ```
#!/bin/bash
Wazuh - Yara active response
Copyright (C) 2015-2022, Wazuh Inc.
#
This program is free software; you can redistribute it
and/or modify it under the terms of the GNU General Public
License (version 2) as published by the FSF - Free Software
Foundation.

#----- Gather parameters -----#

Extra arguments
read INPUT_JSON
YARA_PATH=$(echo $INPUT_JSON | jq -r .parameters.extra_args[1])
YARA_RULES=$(echo $INPUT_JSON | jq -r .parameters.extra_args[3])
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.syscheck.path)
```

```
Set LOG_FILE path
LOG_FILE="logs/active-responses.log"

size=0
actual_size=$(stat -c %s ${FILENAME})
while [${size} -ne ${actual_size}]; do
 sleep 1
 size=${actual_size}
 actual_size=$(stat -c %s ${FILENAME})
done

#----- Analyze parameters -----#

if [[! $YARA_PATH]] || [[! $YARA_RULES]]
then
 echo "wazuh-yara: ERROR - Yara active response error. Yara path and rules parameters are mandatory." >> ${LOG_FILE}
 exit 1
fi

#----- Main workflow -----#

Execute Yara scan on the specified filename
yara_output=$(("${YARA_PATH}"/yara -w -r "${YARA_RULES}" "${FILENAME}")")

if [[$yara_output != ""]]
then
 # Iterate every detected rule and append it to the LOG_FILE
 while read -r line; do
 echo "wazuh-yara: INFO - Scan result: $line" >> ${LOG_FILE}
 done <<< "$yara_output"
fi

exit 0;
```

```
root@linux-machine:~# cp /media/yara.sh /var/ossec/active-response/bin/
root@linux-machine:~# ls /var/ossec/active-response/bin/
default-firewall-drop firewall-drop host-deny ipfu kaspersky.py pf restart-wazuh wazuh-slack
disable-account firewall-drop ip-customblock kaspersky npf restart.sh route-null yara.sh
root@linux-machine:~#
```

- Change `yara.sh` file owner to `root:wazuh` and file permissions to `0750` :

```
root@linux-machine:~# chown root:wazuh /var/ossec/active-response/bin/yara.sh
root@linux-machine:~# chmod 750 /var/ossec/active-response/bin/yara.sh
root@linux-machine:~#
```

- Add the following within the `<syscheck>` block of the Wazuh agent `/var/ossec/etc/ossec.conf` configuration file to monitor the directory

```
GNU nano 7.2 /var/ossec/etc/ossec.conf *
<directories realtime="yes">/tmp/yara/malware</directories>
```

- Restart the Wazuh agent

## Configuring Wazuh server

- Add the following rules to the `/var/ossec/etc/rules/local_rules.xml` file. The rules detect FIM events in the monitored directory. They also alert when the YARA integration finds malware. You can modify the rules to detect events from other directories:
  - Copied prev version, then added new groups and copied back

```
<group name="syscheck,">
 <rule id="100300" level="7">
 <if_sid>550</if_sid>
 <field name="file">/tmp/yara/malware/</field>
 <description>File modified in /tmp/yara/malware/ directory.</description>
 </rule>
```

```

<rule id="100301" level="7">
 <if_sid>554</if_sid>
 <field name="file">/tmp/yara/malware/</field>
 <description>File added to /tmp/yara/malware/ directory.</description>
</rule>
</group>

<group name="yara,">
 <rule id="108000" level="0">
 <decoded_as>yara_decoder</decoded_as>
 <description>Yara grouping rule</description>
 </rule>
 <rule id="108001" level="12">
 <if_sid>108000</if_sid>
 <match>wazuh-yara: INFO - Scan result: </match>
 <description>File "${yara_scanned_file}" is a positive match. Yara rule: ${yara_rule}</description>
 </rule>
</group>

```

```

root@linux-wazuh:~# cp /media/local_rules.xml /var/ossec/etc/rules/local_rules.xml

```

- Add decoders to local `var/ossec/etc/decoders/local_decoder.xml`, so that YARA stuff will be decoded properly
  - Copied prev version, then added new groups and copied back

```

<decoder name="yara_decoder">
 <prematch>wazuh-yara:</prematch>
</decoder>

<decoder name="yara_decoder1">
 <parent>yara_decoder</parent>
 <regex>wazuh-yara: (\S+) - Scan result: (\S+) (\S+)</regex>
 <order>log_type, yara_rule, yara_scanned_file</order>
</decoder>

```

```

root@linux-wazuh:~# cp /var/ossec/etc/decoders/local_decoder.xml /media/local_before_decoders.xml
root@linux-wazuh:~# cp /media/local_decoder.xml /var/ossec/etc/decoders/local_decoder.xml

```

- Add the following configuration to the Wazuh server `var/ossec/etc/ossec.conf` configuration file. This configures the Active Response module to trigger after the rule 100300 and 100301 are fired:
  - Copied prev version, then added new groups and copied back

```

<ossec_config>
 <command>
 <name>yara_linux</name>
 <executable>yara.sh</executable>
 <extra_args>-yara_path /usr/local/bin -yara_rules /tmp/yara/rules/yara_rules.yar</extra_args>
 <timeout_allowed>no</timeout_allowed>
 </command>

 <active-response>
 <disabled>no</disabled>
 <command>yara_linux</command>
 <location>local</location>
 <rules_id>100300,100301</rules_id>
 </active-response>
</ossec_config>

```

```

ossec.conf
319 <ossec_config>
320 <localfile>
322 <location>journald</location>
323 </localfile>
324
325 <localfile>
326 <log_format>syslog</log_format>
327 <location>/var/ossec/logs/active-responses.
 log</location>
328 </localfile>
329
330 <localfile>
331 <log_format>syslog</log_format>
332 <location>/var/log/dpkg.log</location>
333 </localfile>
334
335 </ossec_config>
336
337 <ossec_config>
338 <command>
339 <name>yara_linux</name>
340 <executable>yara.sh</executable>
341 <extra_args>-yara_path /usr/local/bin
 -yara_rules /tmp/yara/rules/yara_rules.yar</
 extra_args>
342 <timeout_allowed>no</timeout_allowed>
343 </command>
344
345 <active-response>
346 <disabled>no</disabled>
347 <command>yara_linux</command>
348 <location>local</location>
349 <rules_id>100300,100301</rules_id>
350 </active-response>
351 </ossec_config>

```

- Restart the Wazuh manager to apply the configuration changes:

```

cess: 14636 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
asks: 219 (limit: 7075)
memory: 471.3M (peak: 489.2M)
CPU: 43.593s
Group: /system.slice/wazuh-manager.service
├─14699 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─14700 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─14701 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─14704 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─14707 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─14748 /var/ossec/bin/wazuh-authd
├─14760 /var/ossec/bin/wazuh-db
├─14771 /var/ossec/bin/wazuh-execd
├─14803 /var/ossec/bin/wazuh-analysisd
├─14813 /var/ossec/bin/wazuh-syscheckd
├─14833 /var/ossec/bin/wazuh-remoted
├─14907 /var/ossec/bin/wazuh-logcollector
├─14932 /var/ossec/bin/wazuh-monitord
└─14943 /var/ossec/bin/wazuh-modulesd

16:34:03 Linux-Mazuh env[14636]: Started wazuh-syscheckd...
16:34:05 Linux-Mazuh env[14636]: Started wazuh-remoted...
16:34:06 Linux-Mazuh env[14636]: Started wazuh-logcollector...
16:34:06 Linux-Mazuh env[14941]: 2026/03/25 16:34:06 wazuh-modulesd:router: INFO: Loaded router module.
16:34:06 Linux-Mazuh env[14941]: 2026/03/25 16:34:06 wazuh-modulesd:content_manager: INFO: Loaded conte
16:34:06 Linux-Mazuh env[14941]: 2026/03/25 16:34:06 wazuh-modulesd:inventory-harvester: INFO: Loaded I
16:34:07 Linux-Mazuh env[14636]: Started wazuh-modulesd...
16:34:09 Linux-Mazuh env[14636]: Completed.
16:34:09 Linux-Mazuh systemd[1]: Started wazuh-manager.service - Wazuh manager.
Linux-Mazuh:~#

```

Rules (4)

From here you can manage your rules.

ID	Description	Groups	Level	File	Path
100000	File modified in /tmp/yara/malware/ directory.	syscheck	7	local_rules.xml	etc/rules
100301	File added to /tmp/yara/malware/ directory.	syscheck	7	local_rules.xml	etc/rules
108000	Yara grouping rule	yara	0	local_rules.xml	etc/rules
108001	File "yara_scanned_file" is a positive match. Yara rule: yara_rule	yara	12	local_rules.xml	etc/rules

Rows per page: 10

Figure — YARA integration configured in SIEM (Rules)

Decoders (2)

From here you can manage your decoders.

Name	Program name	Order	File	Path
yara_decoder			local_decoder.xml	etc/decoders
yara_decoder1	log_type, yara_rule, yara_scanned_file		local_decoder.xml	etc/decoders

Rows per page: 10

Figure — YARA integration configured in SIEM (Rules)

Installing malware and attacking myself with it

- Now I downloaded malware sample and placed inside /tmp/yara/malware

```

root@linux-machine:~# mkdir /tmp/yara/
root@linux-machine:~# mkdir /tmp/yara/malware/
root@linux-machine:~# mkdir /tmp/yara/malware/sus
root@linux-machine:~#

```

```

git clone https://github.com/cado-security/DFIR_Resources_Whispergate
cd DFIR_Resources_Whispergate/Samples
unzip -P "infected" a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92.zip
mv a1* /tmp/yara/malware/sus

```

```

root@linux-machine:~/DFIR_Resources_Whispergate/Samples# cd /tmp/yara/malware/sus/
root@linux-machine:~/tmp/yara/malware/sus# ls
a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92.zip
root@linux-machine:~/tmp/yara/malware/sus# unzip a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92.zip
root@linux-machine:~/tmp/yara/malware/sus# ls
(a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92.2.zip) a196c6b8ffcb97ffb276d04f354696e2391311db3841ae16c8c9f56f36a38e92
inf.txt

```

1,502 hits

timestamp	agent.name	rule.description	rule.level	rule.id
2026 @ 05:58:17.8...	linux-host	File "/tmp/yara/malware/sus" is a positive match. Yara rule: APT_HKTL_Wiper_WhisperGate_Jan22_1_RID331C	12	108001
2026 @ 05:58:17.8...	linux-host	File "/tmp/yara/malware/sus" is a positive match. Yara rule: Whispergate_Stage_1	12	108001
2026 @ 05:58:15.2...	linux-host	File added to /tmp/yara/malware/ directory.	7	100301

Figure — WhisperGate sample detected (no execution)

3.3 Feature 3 — Brute-force attack simulation with attacker user blocking

- For simulating a brute force attack I used hydra
- I initiated the attack on the “Trudy” machine
- Then inside Wazuh dashboard I opened Threat Hunting section and observed remains of unsuccessful acces to my victim machine

```
Example: simulate SSH brute force with Hydra
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://<target-ip> -t 4

Check SIEM rule triggering for brute force (Wazuh rule 5551, 5712, etc.)
```

```
root@Linux-Wazuh:~# hydra -l badguy -P /usr/share/wordlists/rockyou.txt ssh://192.168.30.25 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret ser
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-24 12:59:33
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ssh://192.168.30.25:22/
1 of 1 target completed, 0 valid password found

root@Linux-Wazuh:~# hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.30.25 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-24 12:42:25
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ssh://192.168.30.25:22/
1 of 1 target completed, 0 valid password found
```

Figure — Brute force attack launched from attacker machine

W. Threat Hunting 1 hit

timestamp	agent.name	rule.description	rule.level	rule.id
2026 @ 15:42:28.5...	linux-machine	sshd: authentication failed.	5	5760

Figure — SIEM alert: brute force detected on Device 1

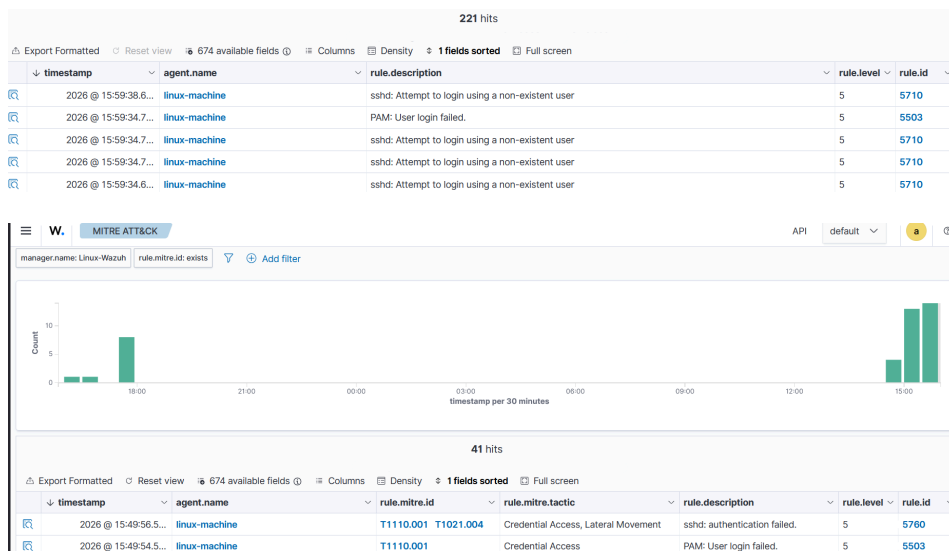


Figure — SIEM alert: brute force detected on Device 1

## Task 4 — Log Management

### 4.1 Log Rotation Setup

Log rotation policy is a system manages logs, alerts, events: defines how long will this data stay in states of live/active/deleted/archived.

Linux has built-in tool for log rotation `logrotate`, providing 2 strategies of how to log rotate:

1. Time-based: delete older then `x` days
2. Size-based: delete based on size and quantity.

This is basic configuration:

```
/var/log/apache2/access_log {
 compress
 dateext
 maxage 365
 rotate 99
 size=+4096k
 notifempty
 missingok
 create 644 root root
 postrotate
 /etc/init.d/apache2 reload
 endscript
}

/var/log/apache2/error_log {
 compress
 dateext
 maxage 365
 rotate 99
 size=+1024k
 notifempty
 missingok
 create 644 root root
 postrotate
 /etc/init.d/apache2 reload
 endscript
}
```

Figure. log config examples

- Explanation of commands:

<code>daily</code>	<code># Rotate logs every day</code>
<code>missingok</code>	<code># Ignore if logs are missing</code>
<code>rotate 14</code>	<code># Keep 14 days of logs</code>
<code>compress</code>	<code># Compress old logs (saves space)</code>
<code>delaycompress</code>	<code># Wait until next rotation to compress</code>
<code>notifempty</code>	<code># Skip rotation if log is empty</code>
<code>create 640 root adm</code>	<code># Set permissions on new log files</code>
<code>sharedscripts</code>	<code># Run post-rotation script once</code>

## Enabling wazuh for log rotations

- I followed this [guide](#) on setting up log rotation for my machine.
  - Created file `/etc/logrotate.d/` and put inside:

```
/var/ossec/logs/alerts.log {
 daily
 rotate 40
 compress
 delaycompress
 missingok
```

```

notifempty
create 0640 root root
sharedscripts
postrotate
 /var/ossec/bin/ossec-logtest -i
 /var/ossec/bin/ossec-control restart
endscript
}

```

- Enabling wazuh archives, inside `/var/ossec/etc/ossec.conf`

```

<ossec_config>
 <global>
 <jsonout_output>yes</jsonout_output>
 <alerts_log>yes</alerts_log>
 <logall>yes</logall>
 <logall_json>yes</logall_json>
 ...
</ossec_config>

```

- Setting up manual time based policy inside `/var/ossec/etc/local_internal_options.conf`

```

GNU nano 7.2 local_internal_options.conf *
local_internal_options.conf
#
This file should be handled with care. It contains
run time modifications that can affect the use
of OSSEC. Only change it if you know what you
are doing. Look first at ossec.conf
for most of the things you want to change.
#
This file will not be overwritten during upgrades.
monitord.keep_log_days: 80

```

Figure. manual time based policy configuration

```

root@vm-honeypot-net:/var/ossec/logs/wazuh/2026/Apr# docker container ls
CONTAINER ID IMAGE COMMAND NAMES CREATED STATUS PORTS
had9b4da91f1 wazuh/wazuh-dashboar... "/entrypoint.sh" single-node-wazuh-dashboar... 5 days ago Up 5 days 443/tcp, 0.0.0.0:443->5681/tcp,
afda89d5e1a wazuh/wazuh-manager:4.10.4 "/init" single-node-wazuh-manager-1 5 days ago Up 5 days 0.0.0.0:1514->1514-1515/tcp,
p. 0.0.0.0:15080->15080/tcp, [::]:15080->15080/tcp, 1516/tcp single-node-wazuh-manager-1 5 days ago Up 5 days 0.0.0.0:9200->9200/tcp, [::]:920
c2b081edc416 wazuh/wazuh-indexer:4.10.4 "/entrypoint.sh open..." single-node-wazuh-indexer-1 5 days ago Up 5 days 0.0.0.0:9200->9200/tcp, [::]:920
1f098ac32ad0 cowrie/cowrie:latest "/cowrie/cowrie-env/..." single-node-wazuh-indexer-1 6 days ago Up 5 days 0.0.0.0:22->2222/tcp, [::]:22->2
cowrie
root@vm-honeypot-net:/var/ossec/logs/wazuh/2026/Apr# docker exec -it afda89d5e1a bash
bash-5.2# ls -la /var/ossec/logs/alerts/
total 116
drwxrwxrwx 3 wazuh wazuh 4096 Apr 19 08:00
drwxrwxrwx 8 wazuh wazuh 4096 Apr 19 08:00 .
drwxrwxrwx 2 wazuh wazuh 4096 Apr 19 18:08 2026
-rw-rw-rw- 2 wazuh wazuh 63323 Apr 19 18:08 alerts.json
-rw-rw-rw- 2 wazuh wazuh 33950 Apr 19 18:08 alerts.log
bash-5.2# tail -f /var/ossec/logs/alerts/alerts.log
User: root
Apr 19 18:08:22 vm-honeypot-net sudo[197371]: pam_unix(sudo:session): session opened for user root(uid=0) by honey-eater(uid=0)
uid: 0
** Alert 1776593303.33569: - pam.syslog.pci_dss.10.2.5.gpg13.7.8.gpg13.7.9.gdpr.IV.32.2.hipaa.164.312.b.nist.800.53_AU.14.nist.8
2026 Apr 19 18:08:23 (vm-honeypot-net) any->journald
Rule: 5582 (level 3) -> PM: Login session closed.
User: root
Apr 19 18:08:22 vm-honeypot-net sudo[197371]: pam_unix(sudo:session): session closed for user root

```

Figure. logs on wazuh-manager on my another machine (not this lab)

- Wazuh stores logs by default on the wazuh-server instead of each agent.
- Wazuh stores logs inside a few possible places. They are retained for 31 days by default
  - `/var/ossec/logs/*`
  - `/var/ossec/logs/alerts/` — alerts
  - `/var/ossec/logs/archives/` — for archived ones/
  - Main place for logs is inside wazuh-manager files json and log format: `/var/ossec/logs/alerts/alerts.log`  
`/var/ossec/logs/alerts/alerts.json`

```

root@vm-honeypot-net:/var/ossec/logs# ls
active-responses.log ossec.log wazuh
root@vm-honeypot-net:/var/ossec/logs# cd wazuh/
root@vm-honeypot-net:/var/ossec/logs/wazuh# ls
2026
root@vm-honeypot-net:/var/ossec/logs/wazuh# cd 2026/
root@vm-honeypot-net:/var/ossec/logs/wazuh/2026# ls
Apr
root@vm-honeypot-net:/var/ossec/logs/wazuh/2026# cd Apr/
root@vm-honeypot-net:/var/ossec/logs/wazuh/2026/Apr# ls
ossec-18.log.gz
root@vm-honeypot-net:/var/ossec/logs/wazuh/2026/Apr#

```

Figure. examples of logs and folder tree of archives logs on my another machine (not this lab)

- The current `alerts.log` and `json` file is hardlinked to the rotated one. Upon rotation tomorrow, a fresh `alerts.log` will be linked to `2026/Apr/ossec-alerts-19.log` and so on.

```

root@vm-honeypot-net:/var/ossec/logs/wazuh/2026/Apr# docker exec -it afda09d5e51a bash
bash-5.2# find / -samefile /var/ossec/logs/alerts/alerts.json 2>/dev/null
/var/ossec/logs/alerts/2026/Apr/ossec-alerts-19.json
/var/ossec/logs/alerts/alerts.json
bash-5.2# find / -samefile /var/ossec/logs/alerts/alerts.log 2>/dev/null
/var/ossec/logs/alerts/2026/Apr/ossec-alerts-19.log
/var/ossec/logs/alerts/alerts.log

```

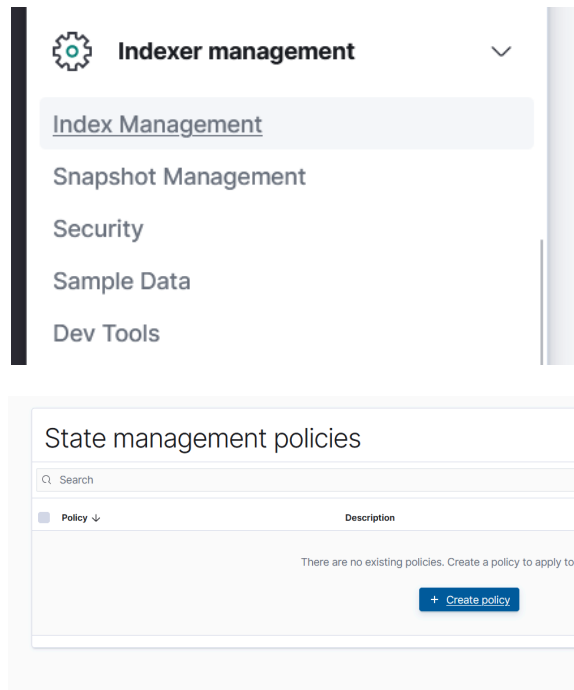
## Indices retention policy in Wazuh

Indices is a low-level storage type format for storing logs and alerts by Wazuh Indexer, provides fast retrieval and access via API or query language like from Kibana and Discovery section of Wazuh.

Indices are not human readable as plain-text or JSON logs.

Wazuh Indexer ingests logs, parses them and then stores them in indices.

- Opening wazuh section:
  - Indexer management → Index Management → State Management Policies → Create Policy



- Added ISM template for `wazuh-alerts-*` events

**ISM templates – optional**  
Specify ISM template patterns that match the index to apply the policy. [Learn more](#)

Index patterns	Priority	
wazuh-alerts-*	1	<a href="#">Remove</a>

[Add template](#)

- Created an **initial** state (no action) and a **delete\_alerts** state (action = **Delete**).

### Create state: delete\_alerts

**State name**  
delete\_alerts

**Order**  
Add after: initial

**Actions**  
Actions are the operations ISM performs when an index is in a c  
= Delete

[+ Add action](#)

**Transitions**  
Transitions define the conditions that need to be met for a state

- Configured the transition from **initial** to **delete\_alerts** when the condition **Minimum index age** is **80d**. Should look like this at the end.

### State: initial

#### Add transition

Transitions define the conditions that need to be met for a state to change. After all actions in the current state are completed, the policy starts checking the conditions for transitions. [Learn more](#)

#### Destination state

If entering a state that does not exist yet then you must create it before creating the policy.

delete\_alerts

#### Condition

Specify the condition needed to be met to transition to the destination state.

Minimum Index Age

#### Minimum index age

The minimum age required to transition to the next state.

80d

- Applied policy and checked for my JSON result

wazuh-alert-retention-policy Edit Delete View JSON

Policy settings		Error notification	Primary term	Last updated
Policy name	wazuh-alert-retention-policy		1	4/10/2024, 13:11 PM
Policy description	A sample description of the policy			
	Sequence number	0		

ISM Templates (1)	
Index patterns	Priority
wazuh-alerts *	1

Rows per page: 10 < 3 >

States (2)	
*Name and conditions on state machines. "actions" are the operations that perform when an index is in a certain state.  *Transitions" define when to move from one state to another. <a href="#">Learn more</a>	
initial	Transitions
delete_alerts	Actions

Figure. Created policy state

View JSON of wazuh-alert-retention-policy

```
{
 "id": "wazuh-alert-retention-policy",
 "seqNo": 0,
 "primaryTerm": 1,
 "policy": {
 "policy_id": "wazuh-alert-retention-policy",
 "description": "A sample description of the policy",
 "last_updated_time": 1776594733135,
 "schema_version": 21,
 "error_notification": null,
 "default_state": "initial",
 "states": [
 {

```

```
{
 "id": "wazuh-alert-retention-policy",
 "seqNo": 0,
 "primaryTerm": 1,
 "policy": {
 "policy_id": "wazuh-alert-retention-policy",
 "description": "A sample description of the policy",
 "last_updated_time": 1776594733135,
 "schema_version": 21,
 "error_notification": null,
 "default_state": "initial",
 "states": [
 {
 "name": "initial",
 "actions": [],
 "transitions": [
 {
 "state_name": "delete_alerts",
 "conditions": {
 "min_index_age": "80d"
 }
 }
]
 },
 {
 "name": "delete_alerts",
 "actions": [
 {

```

```

 "retry": {
 "count": 3,
 "backoff": "exponential",
 "delay": "1m"
 },
 "delete": {}
 },
],
 "transitions": []
},
"ism_template": [
 {
 "index_patterns": [
 "wazuh-alerts-*"
],
 "priority": 1,
 "last_updated_time": 1776594733135
 }
]
}
}

```

## References

1. Install Mikrotik chr on Virtual box [[image vdi](#)] [[guide](#)].
2. Windows QEMU has VNC console type and Windows 7 w/ IE10. Default credentials: IEUser / PasswOrd!
3. Windows 11 installation from GNS3: <https://www.gns3.com/marketplace/appliances/windows-11-dev-env>
4. Official: <https://www.microsoft.com/ru-ru/software-download>



### Fix - To Continue Enter An Admin Username And Password In Windows 11/ 10 || 'Yes' Button Missing

Fix - To Continue Enter An Admin Username And Password In Windows 11/ 10 || 'Yes' Button Missing

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